

A field-measurable regularized-Coulomb sliding law: the $s_N(N)$ master curve, an N_c inversion, and a tidal-admittance ungrounding early-warning

Abdelrahman Saifelden

Alexandria Higher Institute of Engineering and Technology (AIET), Alexandria, Egypt
`Abdulrahman.Saifelden22011411@aiet.edu.eg`

June 20, 2026

Abdelrahman Saifelden (AIET) — Abdulrahman.Saifelden22011411@aiet.edu.eg

Abstract

Ice-stream sliding weakens as the bed approaches flotation, a behaviour large-scale models impose by hand (Joughin, Smith & Schoof 2019 add an ad-hoc near-flotation ramp to a regularized-Coulomb law). We show this weakening is a **closed-form pole** of the regularized-Coulomb law itself and, crucially, that it is **measurable from surface velocity alone**. Solving $\tau_b = CN(u/(u + u_0))^{1/m}$ at the driving stress gives, exactly, the sliding-sensitivity master curve $|s_N|(N) = m/(1 - (N_c/N)^m)$ with $N_c = \tau_a/C$ — a simple pole at the flotation threshold N_c , verified to 1.4×10^{-4} against the numeric s_N . The curve makes three independent surface observables read the **same** law: a lake-drainage step ($du/u = |s_N| \cdot dN/N$), a continuous ocean-thermal-forcing gating slope, and the **tidal admittance** of velocity (whose 2f/1f harmonic ratio diverges toward N_c). A drainage-step population inverts for N_c to **0.2/0.4/1.0 %** at 5/10/20 % amplitude noise; the tidal probe recovers m and the dimensionless flotation proximity $R = (N_c/N)^m$ from velocity alone (answering the “no knowledge of N ” objection), with the tidal forcing amplitude *measured*, not assumed (CATS2008 across the BedMachine grounding zone: median tidal pressure swing 0.30 % of overburden, rising to 0.45 % within 2–4 km of the line). Because the basal-drag stiffness $\partial\tau_b/\partial u \propto (1 - R)^2/R \rightarrow 0$ at N_c , an ice stream nearing ungrounding should show rising variance and lag-1 autocorrelation of surface speed; we **operationalize** the established critical-slowness early-warning (Scheffer et al. 2009; Dakos et al. 2008; Rosier et al. 2021) as the continuous reading $R(t) = (N_c/N)^m \rightarrow 1$ from tidal admittance. Read through the same lens, three real MacAyeal-lake drainages sit far from the N_c pole and show no critical-slowness signature — a correct **true negative** on stable ice (the detectors fire on planted surges/CSD). The contribution is the field-measurable law and its inversion, not a new early-warning concept; absolute s_N remains uncalibrated and the strong test awaits a stream actually nearing flotation.

1. Introduction

The basal sliding law sets how fast an ice stream flows for a given driving stress and effective pressure $N = p_i - p_w$ (Schoof 2005; Gagliardini et al. 2007). Near the grounding line $N \rightarrow 0$ and sliding becomes Coulomb-limited; regularized-Coulomb laws (Joughin, Smith & Schoof 2019;

Tulaczyk et al. 2000; Gudmundsson 2011) capture this but introduce the near-flotation weakening through an *ad hoc* ramp and a fixed velocity scale u_0 . The difficulty is that N is rarely known: it depends on the unobserved subglacial water pressure.

The tools we use, not claim, are the regularized-Coulomb friction law (Schoof 2005; Joughin et al. 2019), the nonlinear tidal (MSf) velocity response of ice streams (Gudmundsson 2011), and critical-slowness-down early-warning theory, *including its ice-sheet application* (Scheffer et al. 2009; Dakos et al. 2008; Rosier et al. 2021; Boers & Rypdal 2021).

What is new, stated against the closest prior work. Joughin, Smith & Schoof (2019) *impose* near-flotation weakening with a hand-tuned ramp; we *derive* it as a closed-form pole $|s_N| = m/(1 - (N_c/N)^m)$ of the regularized-Coulomb law and turn N_c into a **measured** quantity. Rosier et al. (2021) established critical-slowness-down indicators (lag-1 autocorrelation, variance, Kendall- τ) for grounding-line flux ahead of marine-ice-sheet tipping; we do not re-claim the early-warning but give a specific **operational route** — reading $R(t) = (N_c/N)^m \rightarrow 1$ from the continuous tidal admittance of *surface velocity*, which needs no knowledge of N .

2. The $s_N(N)$ master curve

2.1 $s_N(N)$ as the common observable. A lake drainage is an in-situ effective- pressure *step* whose surge amplitude $du/u = |s_N| \cdot (dN/N)$ measures the sliding sensitivity $s_N = d\ln u_b/d\ln N$; ocean thermal forcing lowers N *continuously*, so the gating slope $d\ln u_*/dT$ measures the **same** s_N ($+0.46 \rightarrow +0.62/^\circ C$, direct BedMachine N); ocean tides oscillate N at known amplitude, a third probe (§4). All three read one curve.

2.2 The closed-form master curve. Solving the regularized-Coulomb law $\tau_b = CN(u/(u + u_0))^{1/m}$ at $\tau_b = \tau_d$ gives, exactly, $|s_N|(N) = m/(1 - (N_c/N)^m)$, $N_c = \tau_d/C \rightarrow m$ far from flotation, a **simple pole** at N_c ($\approx N_c/(N - N_c)$). Verified to 1.4×10^{-4} against the numeric s_N . This *derives* the near-flotation weakening that Joughin, Smith & Schoof (2019) impose by hand (ad hoc $h_{af} < h_T$ ramp + fixed u_0). (validation/synthetic/sn_master_curve.py)

3. Inversion — measuring N_c and m

A drainage-step population recovers the flotation threshold N_c to **0.2 / 0.4 / 1.0 %** at 5 / 10 / 20 % amplitude noise; m is degenerate with the per-event drop far from flotation. The recovery is unbiased and sign-faithful under noise and null under permutation (synthetic plant-and-recover).

4. The tidal third probe — N_c , m , and R from velocity alone

Ocean **tides** give a continuous probe: the fundamental velocity admittance equals $|s_N|$ and the 2f/1f harmonic ratio diverges toward N_c , so admittance + harmonics + the known tidal amplitude recover m and the dimensionless flotation proximity $R = (N_c/N)^m$ **from surface velocity alone** — a sliding-law reading of the Gudmundsson nonlinear MSf response, answering the “no knowledge of N ” objection. The tidal forcing amplitude is **measured**, not assumed: sampling the CATS2008

tide model (USAP-DC 601235) across the BedMachine grounding zone (within 10 km of the line; $n = 1108$ valid cells) gives a tidal ocean-pressure swing with median **0.30 % of ice overburden** (median tide $\eta \approx 0.84m$), rising monotonically from 0.21 % at 8–10 km to 0.45 % at 2–4 km from the grounding line (`validation/synthetic/tidal_admittance_probe.py`, `validation/reports/tidal_forcing_gz.json`).

5. An ungrounding early-warning (operationalizing established CSD theory)

The N_c fold makes the basal drag stiffness $\partial\tau_b/\partial u \propto (1-R)^2/R \rightarrow 0$ (critical slowing down), so an ice stream approaching ungrounding should show **rising variance and lag-1 autocorrelation of its surface speed** (Kendall- $\tau \approx 0.54$ in an OU demonstration with N declining toward N_c). Critical-slowing-down early-warning is established (Scheffer et al. 2009; Dakos et al. 2008) and has already been applied to marine-ice-sheet tipping — Rosier et al. (2021) detected rising lag-1 autocorrelation and variance in grounding-line flux ahead of Pine Island tipping, and Boers & Rypdal (2021) in Greenland surface melt. The contribution here is **not** the early-warning concept but a specific **operational route**: reading $R(t) = (N_c/N)^m \rightarrow 1$ from the continuous tidal admittance of surface velocity. A single-snapshot **spatial** analogue (variance + along-flow correlation length rising toward the grounding line) is also derived (exact Lyapunov solve). (`validation/synthetic/spatial_ews.py`)

The fold exponents and a calibration-free spectral gauge (derived). The warning here is not generic: the s_N pole is the Schoof (2007) grounding-line saddle-node, so the critical-slowing-down exponents are fixed, not fitted. Near $N = N_c$ the restoring eigenvalue $\lambda \propto (1-R)^2/R \sim (N - N_c)^2$, giving $|s_N| \propto (N - N_c)^{-1}$, variance $\propto (N - N_c)^{-2}$ and $AC1 \rightarrow 1$ (recovered exponents $-1.00, -2.00, +2.00$). The same rate has a frequency-domain face: the velocity spectrum is a Lorentzian $S(\omega) = 2D/(\lambda^2 + \omega^2)$ whose corner $f_c = \lambda/2\pi \propto (N - N_c)^2$ *reddens* toward ungrounding (the spectral early-warning of Kuehn 2011; Bury et al. 2020). Crucially the fluctuation–dissipation identity $\text{Var} \cdot 2\pi f_c = D$ holds independent of N , so the noise corner f_c is a **calibration-free** flotation-proximity gauge (it falls as $(N - N_c)^2$ regardless of the unknown noise strength D); combined with the tidal-admittance *response* (§4) it even returns D itself — a response/fluctuation pair from one velocity record. (Derived consistency relations, verified on OU realizations: corner recovered to 3%, FDT product constant to 5×10^{-17} ; `nr3_misi_fold_ews`, `nr8_spectral_fdt_ews`.)

6. The framework on the same real lakes

Reading open CryoSat-2 drainage + ITS_LIVE velocity (3 MacAyeal lakes) through this lens (`validation/external/lake_lag_sn_ews.py`, 5 tests): across the 3 resolved drainage events $|\Delta v/v| \leq 2\%$, mixed sign, near the $\sim 1\%$ year-to-year noise floor — **0 positive $>2\sigma$ surge detections** — so via the master curve these trunk lakes sit **far from the N_c pole**; and **no** lake shows the joint critical-slowing-down signature (rising variance *and* high AC1), so the early-warning correctly returns a **true negative** on stable ice (unit tests confirm the detectors *do* fire on planted surges/CSD, so the null is real). Both probes agree: far from ungrounding.

7. Estimator calibration, scope, and falsification

7.1 Calibration. Every estimator has a synthetic plant-and-recover harness in `pytest`: the master-curve closed form (1.4×10^{-4}), the N_c inversion (0.2/0.4/1.0 %), the tidal admittance/harmonic recovery, and the CSD detectors (fire on planted surges/CSD, null on stable OU).

7.2 Scope and falsification. Absolute s_N is uncalibrated (per-event dN/N is a lumped-storage lower bound); the robust claims are the sign/shape and the *threshold* N_c . With $n \approx 8$ annual points per lake the CSD test is weak; a strong test needs the dense quarterly/GPS series and a stream actually nearing flotation. Falsifiers: a co-located dN +GPS population whose $|s_N(N)|$ does not follow $m/(1 - (N_c/N)^m)$; a stream nearing flotation with no variance/AC1 rise; tidal admittance/harmonics not steepening toward the grounding line.

8. Reproduce

```
pip install -r requirements.txt
python glaciers/validation/synthetic/sn_master_curve.py      # $2 s_N(N) master curve + inversion
python glaciers/validation/synthetic/tidal_admittance_probe.py # $4 tidal third probe
python glaciers/validation/synthetic/spatial_ews.py          # $5 spatial early-warning
python glaciers/validation/external/lake_lag_sn_ews.py        # $6 framework on real lakes
pytest glaciers/tests/test_validation_synthetic.py -v         # $7 estimator calibration
```

Data and code availability

All analysis code and derived products are in the public repository at <https://github.com/abdh0saifelden2-debug/K>, and the results regenerate from the scripts above. The study uses the following open datasets:

- BedMachine Antarctica v4 (Morlighem, 2025, <https://doi.org/10.5067/POJQI54A45HX>; see Morlighem et al., 2020).
- CATS2008 circum-Antarctic tide model: USAP-DC 601235 (Howard et al., 2019, <https://doi.org/10.15784/6012>).
- ITS_LIVE surface velocities (Gardner et al., 2025, <https://doi.org/10.5067/JQ6337239C96>; see Gardner et al., 2018) and CryoSat-2 lake-drainage dates.

BedMachine and ITS_LIVE were obtained from the NASA NSIDC DAAC; CATS2008 from the U.S. Antarctic Program Data Center (<https://www.usap-dc.org>).

References

- Boers, N., & Rypdal, M. (2021). Critical slowing down suggests that the western Greenland Ice Sheet is close to a tipping point. *Proceedings of the National Academy of Sciences* 118(21), e2024192118. doi:10.1073/pnas.2024192118
- Dakos, V., Scheffer, M., van Nes, E. H., Brovkin, V., Petoukhov, V., & Held, H. (2008). Slowing down as an early warning signal for abrupt climate change. *Proceedings of the National Academy of Sciences* 105(38), 14308–14312. doi:10.1073/pnas.0802430105

- Gagliardini, O., Cohen, D., Råback, P., & Zwinger, T. (2007). Finite-element modeling of subglacial cavities and related friction law. *Journal of Geophysical Research: Earth Surface* 112, F02027. doi:10.1029/2006JF000576
- Gardner, A. S., Moholdt, G., Scambos, T., Fahnestock, M., Ligtenberg, S., van den Broeke, M., & Nilsson, J. (2018). Increased West Antarctic and unchanged East Antarctic ice discharge over the last 7 years. *The Cryosphere* 12, 521–547. doi:10.5194/tc-12-521-2018
- Gardner, A., Fahnestock, M., Greene, C. A., Kennedy, J. H., Liukis, M., Lopez, L., & Scambos, T. (2025). MEaSUREs ITS_LIVE Regional Glacier and Ice Sheet Surface Velocities, Version 2 (NSIDC-0776) [Data set]. *NASA NSIDC DAAC*. doi:10.5067/JQ6337239C96
- Gudmundsson, G. H. (2011). Ice-stream response to ocean tides and the form of the basal sliding law. *The Cryosphere* 5, 259–270. doi:10.5194/tc-5-259-2011
- Howard, S. L., Erofeeva, S., & Padman, L. (2019). CATS2008: Circum-Antarctic Tidal Simulation version 2008. *U.S. Antarctic Program Data Center*. doi:10.15784/601235
- Joughin, I., Smith, B. E., & Schoof, C. G. (2019). Regularized Coulomb friction laws for ice sheet sliding: Application to Pine Island Glacier, Antarctica. *Geophysical Research Letters* 46, 4764–4771. doi:10.1029/2019GL082526
- Morlighem, M., Rignot, E., Binder, T., Blankenship, D. D., Drews, R., Eagles, G., et al. (2020). Deep glacial troughs and stabilizing ridges unveiled beneath the margins of the Antarctic ice sheet. *Nature Geoscience* 13, 132–137. doi:10.1038/s41561-019-0510-8
- Morlighem, M. (2025). MEaSUREs BedMachine Antarctica, Version 4 (NSIDC-0756) [Data set]. *NASA National Snow and Ice Data Center DAAC*. doi:10.5067/POJQI54A45HX
- Padman, L., Fricker, H. A., Coleman, R., Howard, S., & Erofeeva, L. (2002). A new tide model for the Antarctic ice shelves and seas. *Annals of Glaciology* 34, 247–254. doi:10.3189/172756402781817752
- Rosier, S. H. R., Reese, R., Donges, J. F., De Rydt, J., Gudmundsson, G. H., & Winkelmann, R. (2021). The tipping points and early warning indicators for Pine Island Glacier, West Antarctica. *The Cryosphere* 15, 1501–1516. doi:10.5194/tc-15-1501-2021
- Scheffer, M., Bascompte, J., Brock, W. A., Brovkin, V., Carpenter, S. R., Dakos, V., Held, H., van Nes, E. H., Rietkerk, M., & Sugihara, G. (2009). Early-warning signals for critical transitions. *Nature* 461, 53–59. doi:10.1038/nature08227
- Schoof, C. (2005). The effect of cavitation on glacier sliding. *Proceedings of the Royal Society A* 461, 609–627. doi:10.1098/rspa.2004.1350
- Siegfried, M. R., & Fricker, H. A. (2021). Illuminating active subglacial lake processes with ICESat-2 laser altimetry. *Geophysical Research Letters* 48, e2020GL091089. doi:10.1029/2020GL091089
- Tulaczyk, S., Kamb, W. B., & Engelhardt, H. F. (2000). Basal mechanics of Ice Stream B, West Antarctica: 1. Till mechanics. *Journal of Geophysical Research: Solid Earth* 105(B1), 463–481. doi:10.1029/1999JB900329